Lecture 9: LLMs for Finance[1]

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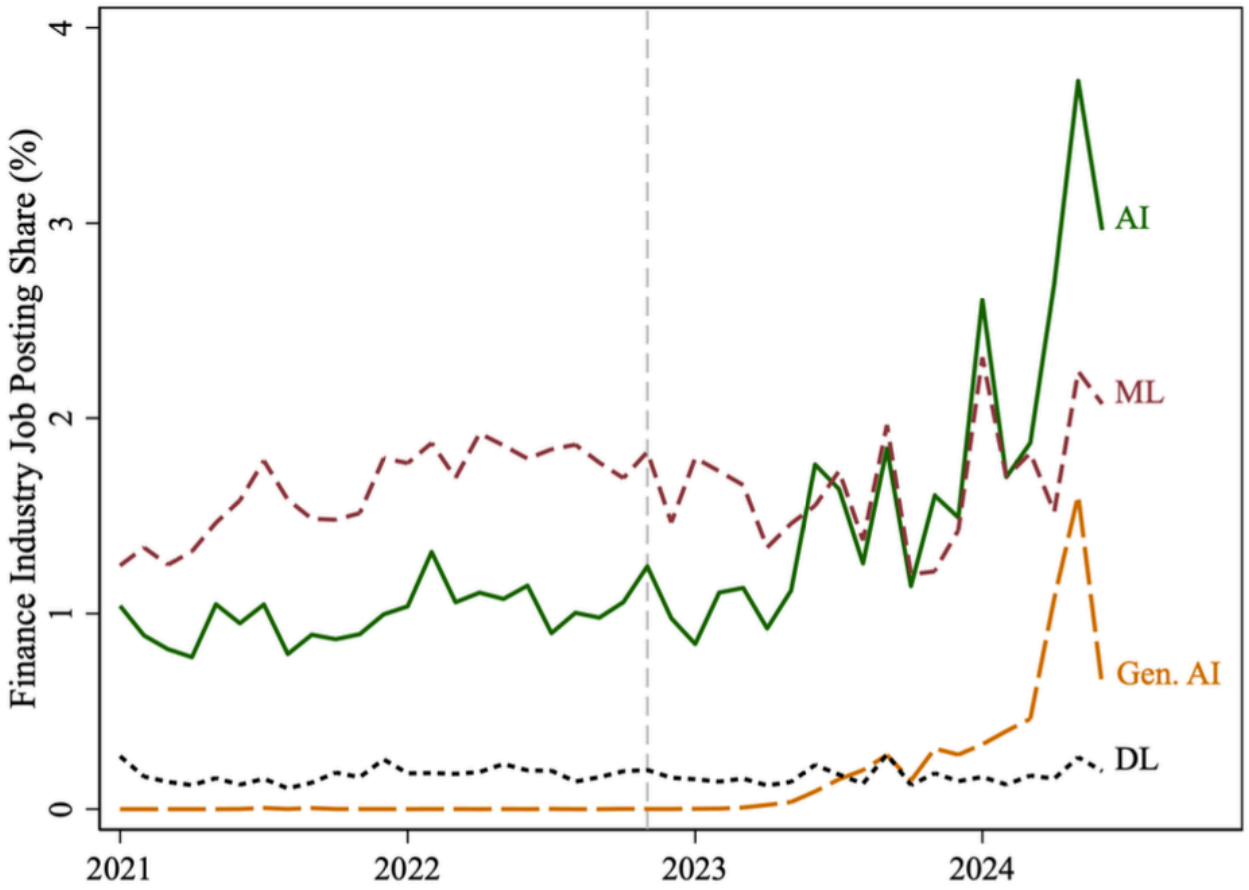
[1] primarily based on Eisfeldt and Schubert [2025]

LLMs and the Financial Industry

LLMs as General Tools for Financial Research

LLM for Investment

Figure 1. Skill demand in the finance and insurance sector. The graph shows the share of online job postings by firms in the finance and insurance sector (NAICS 52) that mention particular skills. "AI" are artificial intelligence skills, "ML" are machine learning skills. "Gen. AI" are Generative AI or large language model skills. "DL" are deep learning skills. The grey drop-down line indicates the ChatGPT release date on November 30, 2022. Data source: Lightcast.



Generative AI impacts firms and the economy

- **A General-Purpose Technology (GPT)**
  - Similar to electricity or the internet: affects **a wide range of tasks**
  - Particularly strong in **information-processing and language-intensive work**
- **Substitution vs Complementarity**
  - *Substitute:* automating routine cognitive tasks (e.g., report writing, summarization)
  - *Complement:* enhancing high-skill decision-making (e.g., research, analysis)
- **Key question**
  - Which financial tasks/jobs/firms are most exposed to LLMs?
  - How to measure the influence?

Generative AI and Firm Values

Key idea: Measure exposure to Generative AI across firms using the task content of their workforce and relate it to stock market reactions to a technology shock.

1. **Measuring jobs' exposure to technology shocks in real time is difficult**
   - Use speed of adoption to provide a real-time assessment in an event-study setting
   - Label tasks based on their exposure to Generative AI productivity changes
2. **Whether a technology substitutes or complements exposed jobs is unclear**
   - If some of your tasks can be replaced by Generative AI, is that good or bad?
   - New approach based on whether an occupation's *core tasks* or *supplemental tasks* are replaceable
3. **Assessing GPT's impact on a firm requires detailed firm-level occupational data**
   - Map from task exposure to occupations and then to firms

Measuring Exposure to Generative AI

- **Occupational Information Network (O*NET):** 19,265 unique tasks mapped to 923 occupations.

Step 1: Task-level exposure

- Assign each task an exposure score based on potential productivity gains from LLMs

$$X_T = \begin{cases} 0 & \text{No exposure} \\ 0.5 & \text{Requires additional infrastructure} \\ 1 & \text{LLM can significantly improve productivity} \end{cases}$$

Step 2: Aggregate to occupations and firms

- Occupation exposure: average task exposure
- Firm exposure: employment-weighted average across occupations

$$X_f = \sum_{O \in f} \text{EmpShare}_{f,O} \cdot X_O$$

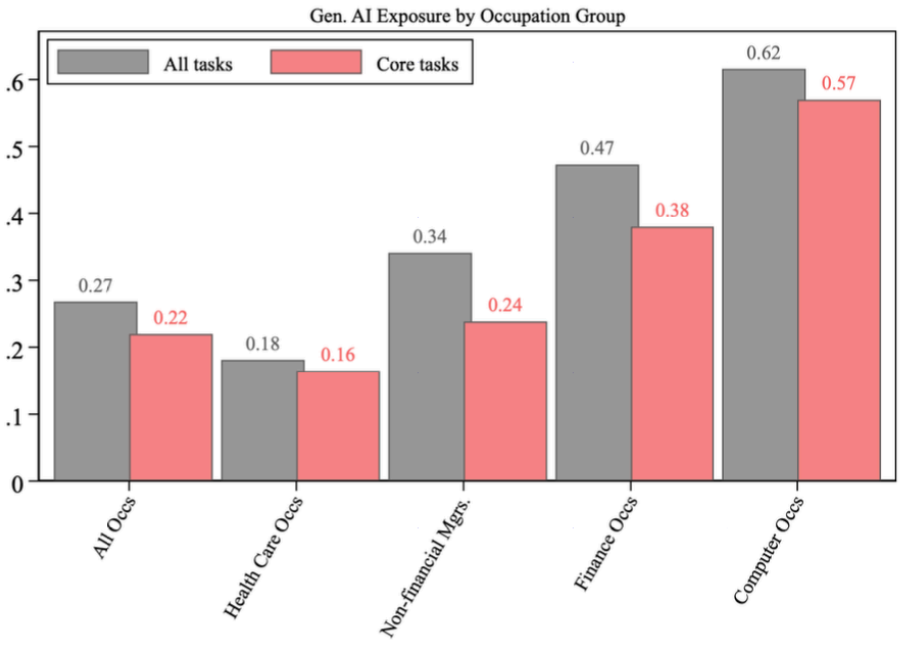
Examples: GPT Scoring of Task Exposure to Generative AI

Occupation	Task	Score	Explanation
Telemarketers	Adjust sales scripts to better target the needs and interests of specific individuals.	1	The model can learn from the sales scripts and the needs and interests of specific individuals to adjust the sales scripts.
Web Developers	Write supporting code for Web applications or Web sites.	1	The model can write code based on the specifications provided in the text input.
Paralegals and Legal Assistants	Investigate facts and law of cases and search pertinent sources, such as public records and internet sources, to determine causes of action and to prepare cases.	.5	The model can assist in searching pertinent sources and summarizing information, but it may not be able to retrieve up-to-date facts from the internet. Additional software could be developed on top of the LLM to retrieve up-to-date facts and combine them with the LLM capabilities.
Heating, Air Conditioning, and Refrigeration Mechanics and Installers	Connect heating or air conditioning equipment to fuel, water, or refrigerant source to form complete circuit.	0	The task requires physical interaction with equipment and materials, which cannot be done by the LLM.
Physician Assistants	Perform therapeutic procedures, such as injections, immunizations, suturing and wound care, and infection management.	0	The task requires physical interaction with patients, which cannot be done by the LLM.

Overall Patterns of Exposure

Average exposure across all occupations: 27%

Panel A: Share of Tasks Exposed to Generative AI Across Selected Occupation Groups



Core vs. supplemental task exposure varies:

- More core exposure in tech roles
- More supplemental exposure in managerial ones

Finance Occupations: High Exposure

Panel B: Share of Tasks Exposed to Generative AI in Finance Occupations

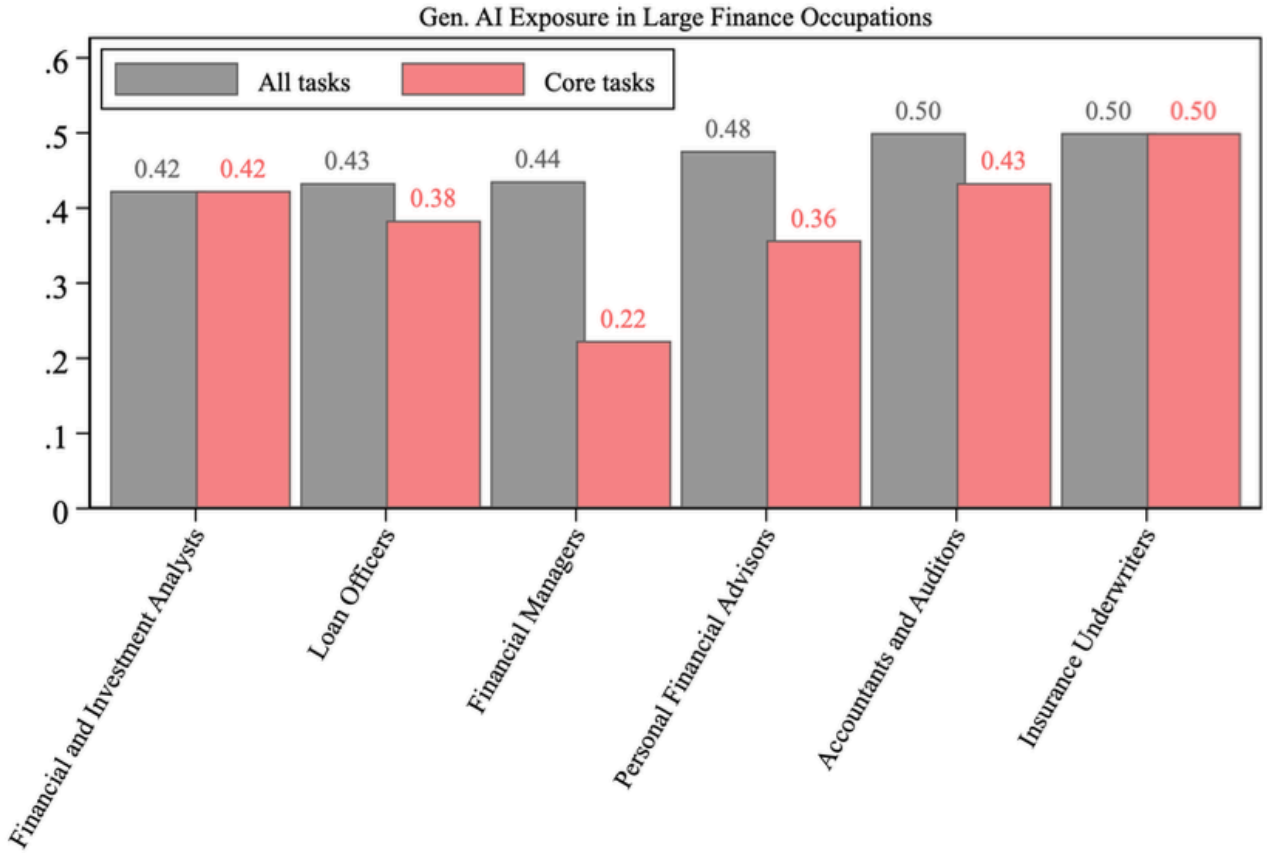


Figure 2. Generative AI Exposure by Occupation. This figure shows the share of tasks in each different white-collar occupation groups exposed to Generative AI productivity improvements (based on the measure in Eisfeldt et al. (2023). Panel A plots the weighted share of tasks exposed to Generative AI in different white-collar occupation groups, where detailed 6-digit occupations are aggregated into weighted averages by occupation groups using data on their relative employment from the BLS. The graph differentiates between overall exposure (in grey) and exposure only from core tasks (in red). Core tasks are defined by the Occupational Information Network as tasks that are both relevant and important for an occupation. Panel

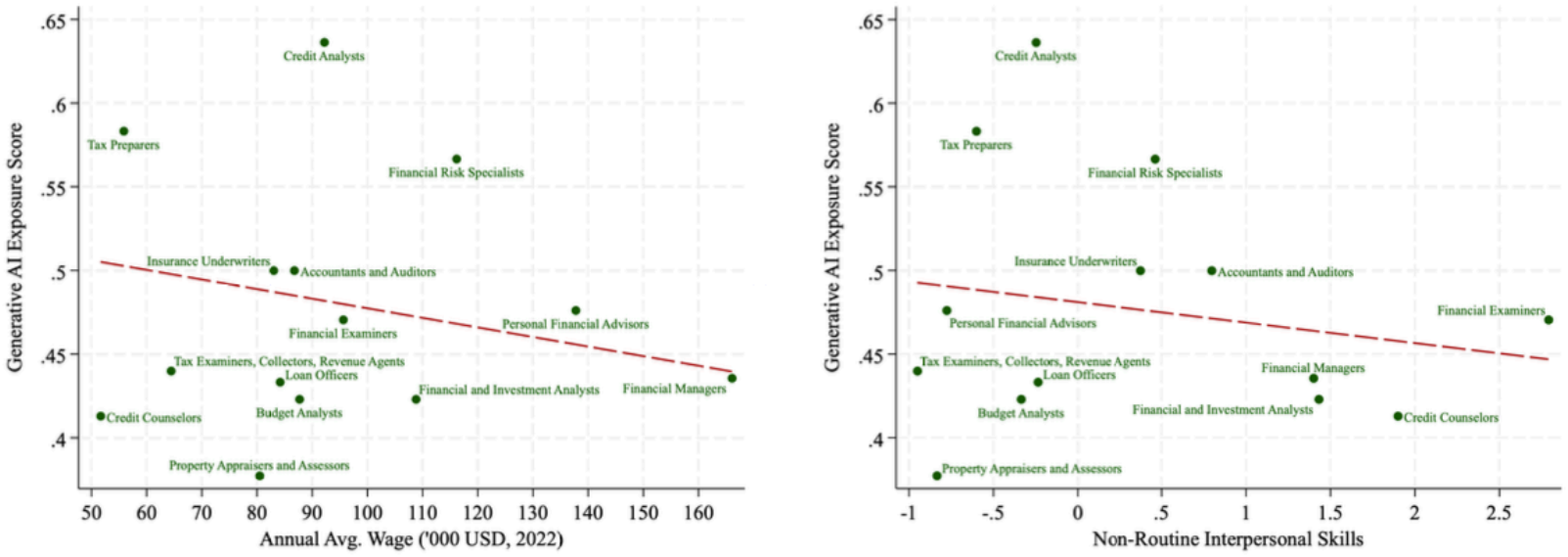
Finance jobs show elevated GenAI exposure:

- Consistent with high average wages ($108K vs. U.S. avg $62K)
- High analytical skill content makes tasks more automatable

Variation within finance:

- Core exposure varies:
  - *Lower core exposure:* Financial managers, loan officers, advisors
  - *Higher core exposure:* Accountants, auditors, underwriters
    - Insurance underwriters review applications for coverage and accept or reject an applicant based on risk analysis.

GenAI Exposure, Wages, and Social Skills



(A) **(B)**

Figure 3. Finance occupation Gen. AI exposure, wages, and social skills. The figures show the relationship between the share of tasks with exposure to Generative AI in different finance occupations and their 2022 wages from the BLS (left panel), as well as the interpersonal skills involved in the occupation (right panel). "Social skills" are proxied by the measure of non-routine cognitive interpersonal skills defined in Acemoglu and Autor (2011) based on O*NET data, which captures to what degree the occupation involves activities like establishing interpersonal relationships, guiding subordinates, or coaching others. Linear fits are weighted by employment.

Within-finance trend: *Higher-wage roles have lower GenAI exposure*

- Opposite to economy-wide pattern, where higher wages $\rightarrow$ higher exposure
- Finance roles already highly analytical

Interpersonal skill content matters:

- Roles needing more *non-routine social skills* show lower exposure
- Examples: Financial analysts, credit counselors

*(Social skills measured via O*NET-based non-routine cognitive interpersonal index)*

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Research Applications of Generative AI

- **Embeddings** for dimensionality reduction and similarity mapping
- **Text classification** with LLMs or fine-tuned transformers
- **Retrieval-Augmented Generation (RAG)** for large corpora
- **Simulating agent behavior** and "synthetic" pilot surveys
- **Hypothesis generation** in creative or early-stage projects

Embeddings

- **Definition:** Dense vector representations of text (or other data) in a lower-dimensional space, trained from contextual prediction tasks
- **Motivation:** Capture semantic relationships, enabling clustering and similarity mapping

Examples

- Chen and Sarkar [2020]: embeddings of 10-K filings to cluster firms
- Gabaix et al. [2025]: "asset embeddings" to cluster assets based on fund holdings and predict co-movement

Embedding Example

Kim et al. [2024] showed the vector-space embedding of heterogeneous labor market entities.

Transformer-Based Text Classification

- **BERT and embedding models**
  - Encoder-based models pre-trained on large corpora
  - Fine-tune on application-specific data for classification tasks
- **Direct classification with an LLM**
  - Zero/few-shot classification on text blocks (e.g. 10-K, earnings calls)
  - Eliminates the need for large labeled training sets, but reliability must be checked

Example: Classification Workflow with GPT-4o API

Goal: Classify segments of 10-K filings for mentions of ESG issues.

1. **Data preparation**
   - Download 10-K text from EDGAR
   - Clean HTML and remove tables / boilerplate
   - Split into smaller chunks (e.g. 800–2000 tokens)
2. **Prompt construction**
   - Example instruction: "Rate each chunk from 0 to 3 for environmental impact mention"
   - Provide definitions and examples for each score
3. **LLM inference**
   - Submit chunks via API in batch mode
   - Enforce stable output format, e.g. a single integer in JSON
4. **Validation**
   - Manually label a random sample
   - Calculate accuracy and confusion matrix; inspect systematic errors

Output: Chunk labels aggregated to firm-level ESG mention intensity.

Pitfalls #1: Text Classification

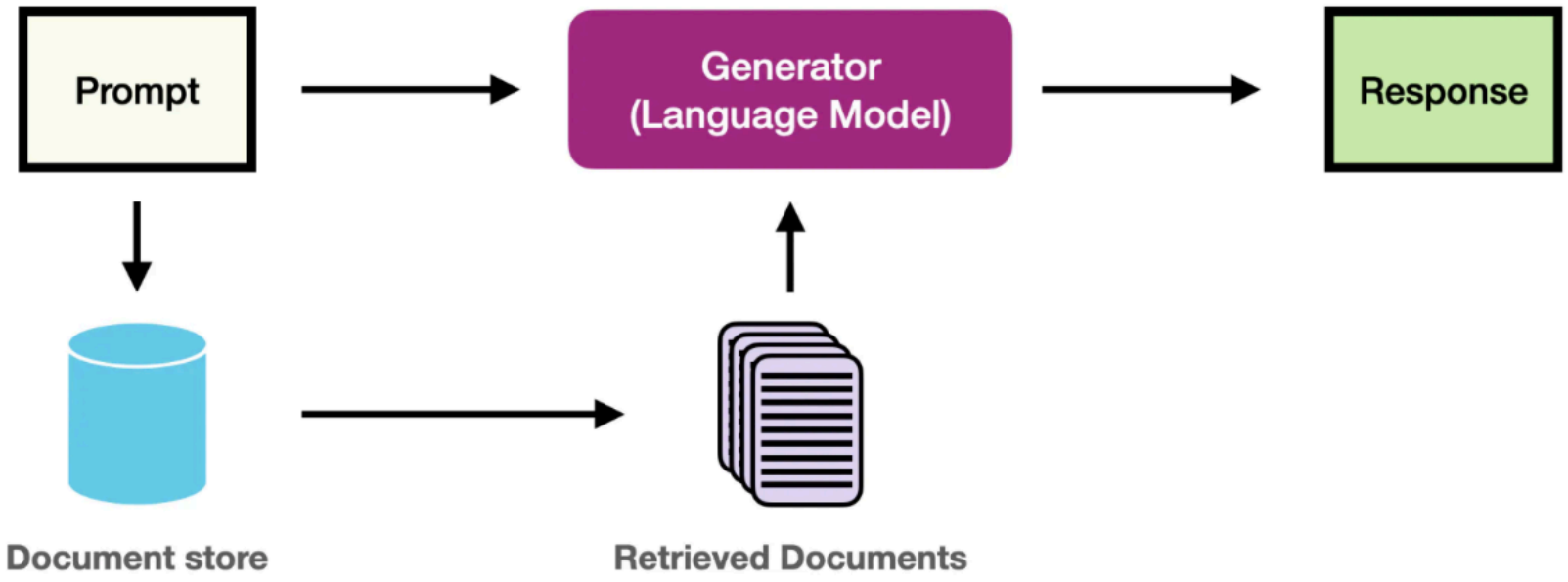
- Large variance or errors may arise for very similar texts
- Ambiguous definitions create unstable labels:
  - "positive relative to what?"
  - "what does 7/10 negativity mean?"
- Numerical scores may not be comparable across texts or runs
- Danger of "model hallucination" or excessive eagerness

Mitigations

- Provide explicit rubrics
- Use few-shot examples
- Check the output distribution
- Demand textual evidence
- Tell the model to rely only on the provided text
- Chain-of-thought: extract evidence first, then classify

RAG: Retrieval-Augmented Generation

- LLMs do not have access to your dataset by default.
- We can add supporting info that the model can draw upon:
  1. Split texts into chunks, embed them, store in vector database.
  2. Retrieves semantically relevant chunks for each query (e.g. using embedding similarity)
  3. Prompt an LLM with these chunks for final classification or summarization.
- RAG allows LLMs to answer questions using external, up-to-date, and domain-specific data.



Pitfalls #2: RAG

- RAG cannot answer overview questions ("how many times does X appear")
- **Chunking**
  - Too coarse: may exceed token limits
  - Too fine: context becomes fragmented
- **Retrieval**
  - Embeddings may retrieve irrelevant or partial chunks
  - Too many weakly relevant chunks can distract the model

Practical fixes

- Semantic splitting using headings or paragraph structure
- Overlapping chunks
- Retrieval quality checks
- Reranking and keyword filtering
- Auditing by saving retrieved chunks together with outputs

Simulating Agent Behavior & Expectations

- **LLM as survey respondent**
  - Rapid, low-cost alternative to pilot surveys
  - Useful for strategic interactions that are hard to model
  - May replicate human biases if prompted with personas
- **Macroeconomic and behavioral studies**
  - LLM-based macro forecasts can resemble the Survey of Professional Forecasters
  - Potential for exploring distributional or demographic heterogeneity (by varying prompt).
- **Caution**
  - Behavior varies by model version and context
  - "The model" does not have intrinsic views; output depends on the prompted persona
  - Always calibrate against real human data

Hypothesis Generation

- **LLMs as idea engines**
  - Brainstorm research directions and alternative explanations
  - Refine concepts through iterative feedback
  - Generate hypotheses from unstructured data
- **Hybrid AI–human teams**
  - Combine human domain expertise with LLM creativity
  - **LLMs are not always good judges of novelty**
  - Validation is often easier for experts than generation
  - Default "second eye" for everything: writing, proofs, empirical approach...

Writing LLM Papers: Validation & Verification

- **Sampling and ground-truth checks**
  - Manually score a random subset and compare with model outputs
  - Essential for text classification and retrieval tasks
- **Prompt engineering**
  - Provide clear definitions, rubrics, and examples
  - Avoid ambiguous requests
  - Document and justify prompt choices
- **Replicability**
  - Document exact model version
  - Fix temperature or seed when possible
  - Report chunk sizes and filtering thresholds for RAG
  - Check sensitivity to small perturbations
  - Clarify numerical scales and category definitions

Writing LLM Papers: Interpretation & Reporting

- **Explain or justify reasoning**
  - Ask for reasoning cautiously
  - Check consistency between scores and reasoning
  - LLMs cannot explain "their" capabilities in any deep sense
- **Beware of biases and hallucinations**
  - Models may rationalize after the fact
  - Conduct repeated tests and sensitivity checks
- **Clarity for reviewers**
  - Provide full prompts and labeled text examples in appendices
  - Report distributional checks, such as histograms of scores and comparisons with ground truth

Summary

Applications of Generative AI in Finance Research

Application	Question Types	Examples
Embeddings	• How to represent complex data concisely? • What are semantic relationships in data? • How to cluster similar entities?	• Gabaix et al. (2023): Asset embeddings • Chen and Sarkar (2020): 10-K filings embeddings • Kim et al. (2024): Labor market clusters
Text Classification	• What is the sentiment of financial text? • How to categorize documents? • What topics are discussed?	• Chang et al. (2024): Earnings call sentiment • Krockenberger et al. (2024): Covenant violations • Caragea et al. (2020): FinTech patent classification
Retrieval-Augmented Generation (RAG)	• How to find relevant information in large datasets? • How to use LLMs for classification based on many potential sources? • How to retrieve similar documents from a corpus?	• Bartik et al. (2023): Housing regulation classification • Chen and Wang (2024): AI patent assingment to functionalities
Simulating Agent Behavior	• Can LLMs replicate human preference heterogeneity? • What are expected survey responses? • What would human expectations be in counterfactual scenarios?	• Fedyk et al. (2024): Asset class preferences • Bybee (2023): Macroeconomic expectations • Hewitt et al. (2024): Experimental outcome predictions
Hypothesis Generation	• How to generate new research ideas or business ideas? • How to conduct qualitative research at scale?	• Si et al. (2024): Novel research idea generation • Girotra et al. (2023): New product ideas • Ludwig and Mullainathan (2024): Feature importance in neural network predictions
Teaching Finance	• How to create engaging course materials? • How to use LLMs for course management? • How to design interactive simulations?	• Mollick et al. (2024): Interactive simulations • Exam preparation and grading schemes • Chatbots for course material review

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Literature Review

- Representation: Chen et al. [2022, 2024]
- Sentiment: Lopez-Lira and Tang [2023]
- Decision agent: Perlin et al. [2025], Li et al. [2025]
- Alpha Mining: Wang et al. [2025]
- Memory / Leakage

Q&A

- Questions or thoughts about today's lecture?

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